
Design Document for **Pocket Deck**

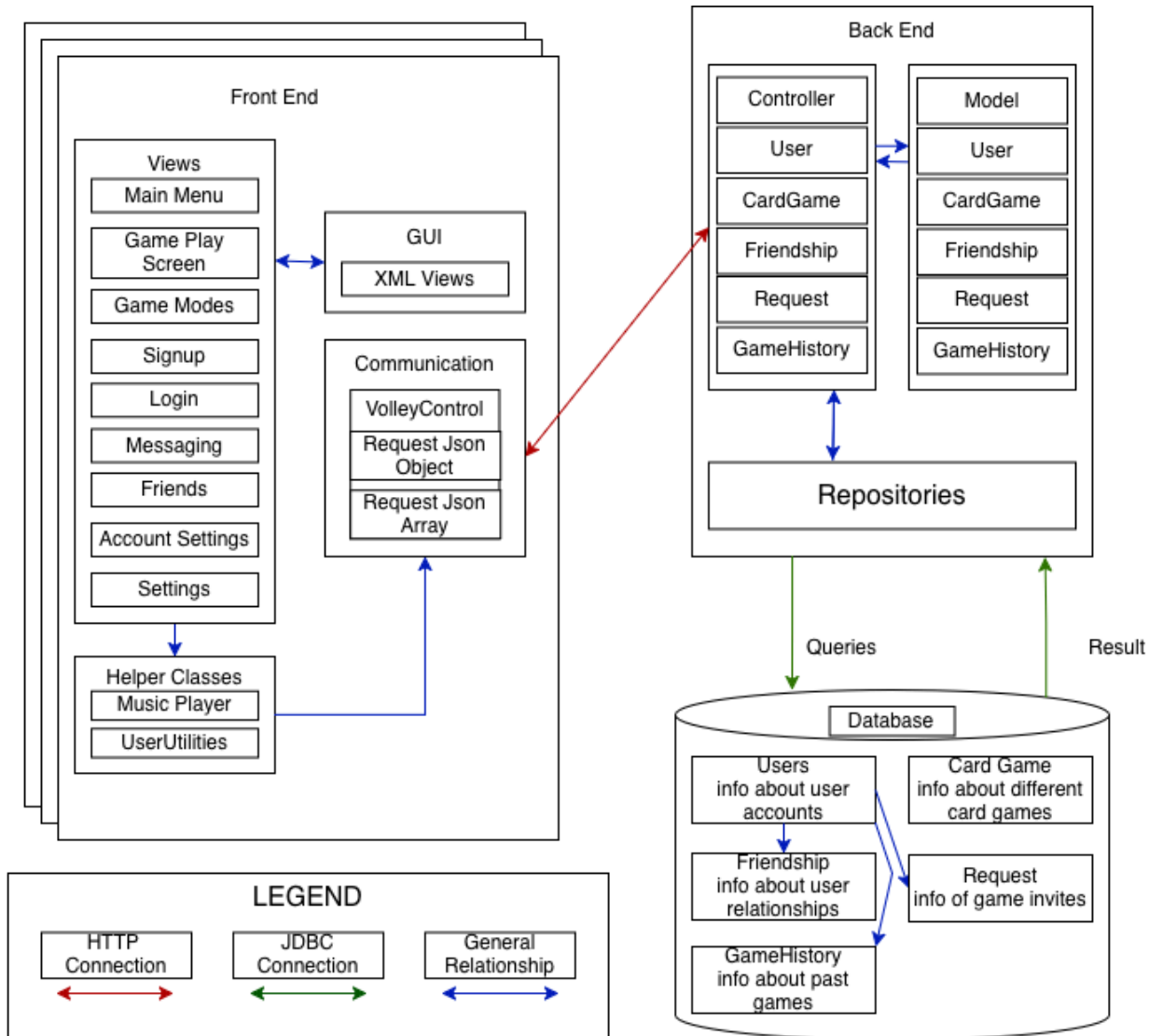
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Frontend (Currently Implemented)

Login

- Signup creates a user object that is stored in the database with username and password elements.
- The login screen allows users to input these values and send them to the backend for accreditation. After being accredited, the user is returned to the main menu and able to select game mode and play games.

Signup

- Allows the user to input a new user with a unique username and password, the latter they must enter successfully twice.
- Passes the new user to the server if all checks pass, handling either the success or failure of writing the new user.

Main Menu

- The main menu creates a page with buttons and menus to serve as a center for all information. The main menu contains the play button, links to login/signup, settings, game modes, game history, and account settings.

Music Player

- MediaPlayer creates a MediaPlayer that is assigned an MP4 file to play music across the application.
- The music player allows for music to be played across the app and also serves as the place where music settings are stored in the userPreferences file. These settings include musicOn and volume. This allows the music setting to be shared across all screens and saved for the user after closing the app or logging out.

Account Settings

- Allows the user to access and edit their user data. This includes their display name and password.
- Page also allows the user to not only log out of their account, but also to delete it from the database entirely if they so desire.

User Utilities

- A helper class containing most of the operations involving the user. Allows for the fetching of username and password, along with a consistent method of encoding user data to send to the server.

Backend

Controllers

The controllers contain the mappings for communication between frontend and the database.

These include:

- User: Contains the above mappings to create, update and delete users, which contain one-to-many relationships with Friendship, Request, and GameHistory.
- CardGame: This adds card game-specific information that is not used much yet.
- Friendship: This has two many-to-one relationships with Users. Each user can send multiple friend requests and receive multiple friend requests. These requests have a status that can be changed, and when a user accepts a friend request, the date is recorded.
- Request: This has two many-to-one relationships with Users. Each user can make multiple requests and receive multiple requests that have a status.
- GameHistory: This has a many to one relationship with Users. Each user can have multiple game history records tied to them. Records will be initialized on creation of a game session, and data will be finalized at the end of the game session.

